TEXORA · LECTURE SHEET

Concept of E-Textiles

Textiles that sense, compute and respond: how microcontrollers, sensors, actuators, LEDs and even solar cells are embedded into garments and textile substrates, the conductive yarns that connect them, and the power, washing and durability problems that still limit them.

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 7 · LEVEL 2 TERM 2

1. What e-textiles are

E-textiles (electronic textiles), also called smart textiles in their wider family, are fabrics and garments that incorporate electronic components and interconnections so that the textile itself can sense conditions, process signals and sometimes respond. Three generations are commonly distinguished:

- **First generation:** conventional electronic devices are attached to or carried in a garment — a sensor pod clipped into a pocket. The textile is only a carrier.
- **Second generation:** the electronics are integrated into the fabric structure — conductive yarns are woven or knitted in as circuits, and textile sensors (for strain, pressure, temperature) are formed by the yarns themselves.
- **Third generation:** the fibre itself is functional — yarns that are transistors, batteries or light-emitters. This is active research rather than everyday manufacture.

Related terms: a **smart textile** in the broad sense reacts to its environment (including non-electronic responses such as phase-change temperature control); **wearable technology** covers devices worn on the body, many of which are not textile at all (watches, wristbands). E-textiles are where the two ideas overlap: the garment *is* the device.

2. The building blocks inside a garment

COMPONENT	ROLE IN AN E-TEXTILE
Microcontroller	The garment's processor: a small, low-power chip (often on a sewable board with large connection pads) that reads the sensors, runs the program, drives the outputs and manages power. It is usually removable for washing.

Sensors	The devices of Topic 6 in textile form: temperature sensors, heart-rate and respiration sensors (electrodes or optical), accelerometers for movement and posture, strain and pressure sensors formed by conductive knitted structures, moisture sensors.
Actuators	Components that act on the wearer or the environment: vibration motors for alerts, shape-memory alloy wires that change a garment's fit or open vents, electroactive materials, and heating elements driven through conductive yarn.
ICs (integrated circuits)	Beyond the controller: amplifier and analogue-to-digital converter chips for signal conditioning, memory, and radio chips (Bluetooth Low Energy is common) that send the garment's data to a phone.
LEDs and displays	Light-emitting diodes sewn or bonded onto the fabric for signalling, visibility (safety clothing) or fashion; fibre-optic strands can spread the light through the cloth.
Transistors and diodes	Present both as packaged components on flexible circuits and, in research, as fibre-level devices; practically, they switch and rectify in the garment's small circuits, exactly as in Topic 5.
Solar cells	Flexible photovoltaic strips or fibres mounted on shoulders, bags or outerwear; they trickle-charge the battery from daylight. Output is small, so they extend rather than replace the battery.
Power source	Usually a thin rechargeable lithium cell, or coin cells for simple circuits — see Section 4.

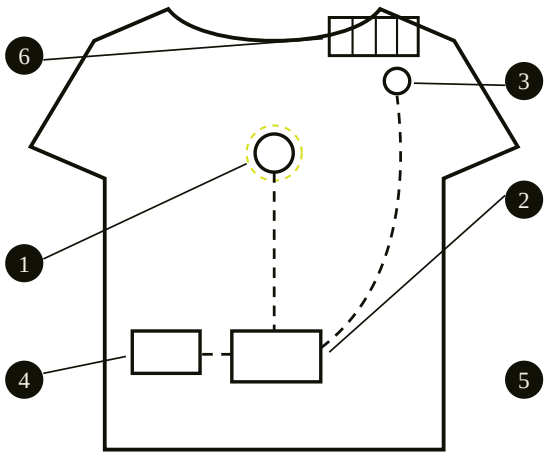


Fig. 1 — Schematic of an e-textile garment. 1 Sensor (chest) · 2 Microcontroller · 3 LED · 4 Thin battery · 5 Conductive-thread tracks (dashed) · 6 Solar cell (shoulder).

3. Interconnections: conductive yarns and textile circuits

What makes an e-textile a textile is that its “wiring” is yarn. **Conductive yarns** are made by coating fibres with metal (silver-coated nylon is common), by wrapping a fine metal filament around a textile core, by using stainless-steel staple fibres blended into the yarn, or with carbon-based conductors. Such yarns can be sewn (including by ordinary embroidery machines, which “print” circuit tracks in thread), woven or knitted into the garment.

- **Embroidered and sewn circuits:** conductive thread stitched in the pattern of a circuit board’s tracks; components are soldered or crimped to the thread, or attached with snap fasteners that double as connectors.
- **Woven and knitted structures:** conductive yarns inserted in warp or weft, or in knitted courses, forming buses, electrodes and resistive sensors directly in the cloth.
- **Printed electronics:** conductive inks screen-printed onto fabric — cheap in quantity, but the print must survive stretching and washing.

Design constraints differ sharply from a printed circuit board: thread resistance is relatively high and changes with stretching and moisture; crossovers must be insulated so tracks do not short when the cloth folds; and every connection is a flexible joint that will be flexed thousands of times. Circuits are therefore kept short, currents small, and critical links doubled.

4. The power-supply problem

Power is the hardest constraint in e-textiles. A garment must stay light, soft and washable, but electronics need energy:

- **Batteries:** coin cells suit simple, low-current circuits (a few LEDs, a sensor that sleeps most of the time). Richer functions need thin rechargeable lithium-polymer cells — which add stiffness and weight, must be protected from bending and puncture, and are normally removed before washing.
- **Energy harvesting:** flexible solar cells (Section 2) recover some energy outdoors; body-heat (thermoelectric) and movement (piezoelectric or triboelectric) harvesters exist but give very small power. Harvesting extends battery life; it rarely replaces the battery.
- **Power management:** the microcontroller sleeps between measurements, sensors are switched on only when read, and radios transmit in short bursts — most e-textile design effort goes into making a small battery last.

5. Washing, durability and other practical issues

- **Washing:** water, detergent, heat and mechanical action attack conductors and joints alike. Solutions include encapsulating components in waterproof pods or coatings, watertight connectors, and designing the electronics as a removable module — still the commonest commercial approach.
- **Mechanical durability:** repeated bending, stretching and abrasion fatigue metal-coated yarns and crack printed tracks; placing circuits away from high-strain zones (elbows, knees) and using stretchable conductor layouts helps.
- **Comfort and safety:** components must not create pressure points or sharp edges; voltages are kept low (a few volts), currents limited, and materials in skin contact must be biocompatible and sweat-resistant.
- **Manufacture and repair:** mixing garment and electronics production lines is organisationally hard, testing is slower than for ordinary garments, and a failed component is difficult to replace once integrated — all reasons e-textiles remain a premium, specialist product.
- **End of life:** electronics inside clothing complicate textile recycling; separating the electronic module helps both reuse and disposal.

6. Example applications

- **Health monitoring:** shirts with textile electrodes recording heart activity and respiration, used in patient monitoring, rehabilitation and elderly care, where a comfortable garment is worn far longer than strapped-on devices.
- **Sportswear:** garments logging heart rate, breathing and movement for training analysis, replacing chest straps with sensing built into the shirt.
- **Heating garments:** jackets and gloves with conductive-yarn heating panels and a battery controller, used by outdoor workers and in cold-storage work.
- **Safety clothing:** high-visibility garments with integrated LED signalling, and workwear that senses posture or exposure.
- **Interactive and fashion uses:** garments whose colour, light or sound responds to touch or environment — the visible showcase of the technology.
- **Military and space:** uniforms monitoring the wearer’s condition and position — historically an important driver of the research (kept general here; specific systems are not public in detail).

Notice the pattern: every application above is a chain of the earlier topics — a transducer (Topic 6) sensing, a diode or transistor circuit (Topic 5) conditioning, a controller deciding, and an actuator or radio responding, all powered and protected with the care of Topics 3 and 4 — but built from yarn.

7. Key terms

- **E-textile:** a textile or garment with electronic components and interconnections built into it, able to sense, process and respond.
- **Conductive yarn:** yarn made electrically conductive by metal coating, metal wrapping or conductive fibres; used as the circuit tracks and electrodes of an e-textile.
- **Wearable technology:** electronic devices designed to be worn on the body; e-textiles are the garment-integrated branch of it.
- **Energy harvesting:** recovering small amounts of energy from light, heat or movement to supplement a battery.
- **Encapsulation:** sealing electronic parts in a waterproof, protective coating or pod so the garment can be washed.
- **Actuator:** a component that produces an action — motion, heat, light or vibration — in response to an electrical signal.

8. Key points

- An e-textile integrates microcontroller, sensors, actuators, ICs, LEDs and a power source into a garment, connected by conductive yarns rather than wires.
- Conductive yarns (silver-coated, metal-wrapped, stainless-steel blends) can be sewn, embroidered, woven or knitted into circuit tracks, electrodes and sensors.
- Power is the main constraint: small batteries, aggressive sleep modes, and solar or other harvesting as a supplement.
- Washing and flexing are the main durability threats; encapsulation and removable electronics modules are the current answers.
- Established application areas are health monitoring, sportswear, heating garments and safety clothing.

9. Summary

E-textiles carry the whole of this course into the fabric itself: the sensors of Topic 6 become knitted electrodes and strain-sensing yarns, the diodes, transistors and amplifiers of Topic 5 shrink onto sewable boards, and the power-supply and protection thinking of Topics 3 and 4 reappears as battery management and encapsulation. The enabling material is the conductive yarn, which lets looms, knitting machines and embroidery machines build circuits as they build cloth. Power supply, washability and durability remain the limiting problems — which is why e-textiles today are specialist, high-value products, and why they are a natural growth area for textile engineers who also understand electronics.

10. Practice

Answer in your own words; brief model answers follow. No multiple-choice questions.

1. List the components you would need for a sports shirt that records heart rate and sends it to a phone, and state the job of each in one line.
2. Give three ways of forming the circuit tracks of an e-textile, and one weakness of each.
3. Why is the power source usually designed to be removable, even though that makes the garment less “integrated”?
4. An LED jacket is to be sold for night-time road work. Identify two durability problems it will face and a design answer to each.
5. Explain why conductive-thread resistance matters more in an e-textile circuit than copper-track resistance on a circuit board.

Answer key

1. Textile electrodes or an optical heart-rate sensor (sensing); a signal-conditioning IC/amplifier (cleaning the small signal); a low-power microcontroller (sampling and processing); a Bluetooth Low Energy radio, often inside the same module (sending data to the phone); conductive yarns (connecting sensor to module); a thin rechargeable battery (power), with a snap or pocket mounting so it can be removed for washing.
2. Sewing/embroidery with conductive thread — tracks can fray, and thread resistance is relatively high. Weaving/knitting conductive yarns into the cloth — the circuit is fixed at manufacture and hard to modify or repair. Screen-printing conductive ink — printed tracks crack under repeated stretching and washing.
3. Because rechargeable cells are damaged by water, heat and the mechanical action of washing, and a punctured or bent lithium cell is a safety risk. A removable pod keeps the expensive electronics safe, lets the garment be washed normally, and lets a flat battery be swapped instead of discarding the garment.
4. (Any two:) Washing attacks the LED connections — encapsulate the LEDs and use a removable battery/controller module. Flexing at shoulders and elbows fatigues the conductive tracks — route tracks away from high-strain zones and use layouts that tolerate stretch. Sweat corrodes metal-coated yarn — use corrosion-resistant (e.g. stainless or well-coated) conductors and sealed joints.
5. Conductive thread has a much higher resistance per metre than a copper track, and that resistance changes as the garment stretches, folds and absorbs moisture. Voltages therefore drop along the thread and sensor readings can shift, so e-textile circuits must use short paths, small currents and generous design margins, unlike board circuits where track resistance is usually negligible.